

Deep Deep Learning

Graduate student-led research seminar | Semester: TBD | Course number: TBD

Instructor: Assaf Shocher. Course contact and office hours: TBD.

Course description

Major ideas in deep learning, studied through foundational material, a central paper and recent developments. The candidate pool includes NTK, Mamba, in-context learning, test-time training, hypernetworks, world models, model merging, interpretability, new generative frameworks, graph neural networks, KANs, and the theory of why training works.

Topic selection

The 21 topics in this syllabus are candidates, not weekly assignments. Students and the instructor choose which topics enter the final 13-week schedule. Unchosen topics are not assigned. Selection, order and presenter assignments: TBD. Each selected topic gets two academic hours.

Learning objectives

Explain a mathematical mechanism; identify its assumptions; assess an experiment that tests an explanation; connect recent research to older ideas; formulate a precise unresolved question.

Prerequisites & logistics

Working knowledge of deep learning, linear algebra, probability, calculus and optimization. Formal prerequisites, credits, enrollment, meeting time, room and delivery mode: TBD. Dates are not yet assigned. Flow matching is assumed background from the base course.

Seminar format & assessment

Students lead presentations and discussion. Each session centers on one main paper, with selected foundation and companion material. Prepare one derivation, one decisive experiment and one unresolved question. No exam. Grading components and weights, attendance requirements, written submissions, reading sections and deadlines: TBD.

Research practice & communication

Cite papers, figures, code and other sources; distinguish the authors' claims from your own analysis. Formal academic-integrity and AI-use policies, course forum and accessibility arrangements: TBD.

Weekly schedule

Week numbers only. Topic selections and student presenters are TBD. The candidate pool below is larger than the final course.

Week	Selected topic	Presenter
1	TBD	TBD
2	TBD	TBD
3	TBD	TBD
4	TBD	TBD
5	TBD	TBD
6	TBD	TBD
7	TBD	TBD
8	TBD	TBD
9	TBD	TBD
10	TBD	TBD
11	TBD	TBD
12	TBD	TBD
13	TBD	TBD

Suggested meeting structure

Two academic hours per selected topic. Assuming two 45-minute academic hours: 15 minutes foundations, 35 minutes mechanism and derivation, 15 minutes evidence, 25 minutes discussion. Adjust the split to the local academic-hour convention.

Choosing one focus

TTT can focus on learning as neural memory or distribution-shift adaptation with the original Sun et al. paper. IT3 is supplementary reading. These are alternative focuses, not two required weeks. Drifting and IMLE can share a comparison session on one-step generation, with one method developed in depth. Selective comparisons keep either topic within one meeting.

Candidate topic pool

Choose one focused question and one main paper for each selected session. Background and recent readings supply selective preparation and comparisons. An alternative focus is a choice within the same candidate topic.

Kernels, feature learning & optimization

NTK and feature learning

When can a neural network be understood as a kernel method, and when do its features change?

Background: Lee et al., [Wide Neural Networks of Any Depth Evolve as Linear Models Under Gradient Descent](#) (2019), selected sections on Taylor linearization and squared-loss dynamics. Review Jacobians, Gram matrices and gradient flow.

Main paper: Jacot, Gabriel & Hongler, [Neural Tangent Kernel: Convergence and Generalization in Neural Networks](#) (2018).

Recent development & discussion: Haas et al., [On the Surprising Effectiveness of Large Learning Rates under Standard Width Scaling](#) (2025).

Presentation focus: Derive function-space gradient flow and its solution for squared loss, then compare a finite network with its linearization. Use the recent paper to examine why width, parameterization, learning rate and the choice of loss matter for claims about feature learning.

muP, scaling and hyperparameter transfer

How should initialization and updates scale as networks become wider or deeper?

Background: Yang & Hu, [Tensor Programs IV: Feature Learning in Infinite-Width Neural Networks](#) (2021), selected sections on parameterization and feature learning. Review the NTK linearized regime as contrasting background; a separate NTK seminar is optional.

Main paper: Yang et al., [Tensor Programs V: Tuning Large Neural Networks via Zero-Shot Hyperparameter Transfer](#) (NeurIPS 2021; arXiv 2022).

Recent development & discussion: Dey et al., [Don't Be Lazy: CompleteP Enables Compute-Efficient Deep Transformers](#) (2025).

Optional theory: Chen, Yang, Zhao & Gu, [Global Convergence and Rich Feature Learning in L-Layer Infinite-Width Neural Networks under \$\mu\$ Parametrization](#) (ICML 2025). Read the assumptions and result statement; this infinite-width analysis does not establish unrestricted finite-network SGD convergence.

Presentation focus: Work through activation and update scaling in a small multilayer network, and inspect a learning-rate transfer experiment. Discuss the additional constraints introduced by depth; width transfer alone does not establish depth transfer or effective nonlinear learning.

Geometry of optimization: Muon

How does the norm used to measure an update change the direction of steepest descent?

Background: Bernstein & Newhouse, [Old Optimizer, New Norm: An Anthology](#) (2024), selected sections on steepest descent, dual norms and matrix operator norms.

Main paper: Liu et al., [Muon Is Scalable for LLM Training](#) (2025).

Recent development & discussion: Kovalev, [Understanding Gradient Orthogonalization for Deep Learning via Non-Euclidean Trust-Region Optimization](#) (2025).

Presentation focus: Derive the spectral-norm-constrained linearized update using the SVD and explain its polar factor and Newton-Schulz approximation. Examine a scaling experiment and distinguish the geometric argument from implementation choices, weight decay and empirical comparisons with AdamW.

Why training works: convergence and loss landscapes

Which assumptions let us prove that training reaches a global minimum, and how close are those proofs to practical networks?

Scope for one session: One seminar of two academic hours. Choose a convergence proof for a restricted model (shown below), or the edge-of-stability dynamics focus. A survey of all optimization and generalization theory would be too broad for one meeting.

Background: Karimi, Nutini & Schmidt, [Linear Convergence of Gradient and Proximal-Gradient Methods Under the Polyak-Lojasiewicz Condition](#) (2016), the PL inequality and gradient-flow loss decay. Historical context: Choromanska et al., [The Loss Surfaces of Multilayer Networks](#) (2015; 2014 preprint), the spin-glass analogy and its assumptions. Optional contrast: Soudry & Hoffer, [Exponentially vanishing sub-optimal local minima in multilayer neural networks](#) (ICLR 2018), volumes of regions containing minima versus a training trajectory.

Main paper: Dana, Pillaud-Vivien & Bach, [Convergence of Shallow ReLU Networks on Weakly Interacting Data](#) (NeurIPS 2025). Gradient flow in finite shallow ReLU networks, weakly interacting inputs and a particular asymmetric randomized initialization.

Recent development & discussion: Petit, Poon & Peyré, [On the global convergence of gradient flow for wide shallow models beyond homogeneous nonlinearities](#) (2026). The result is conditional: if the studied Wasserstein gradient flow converges, its limit is globally optimal. Watch list: Soudry's [publication page](#) lists Global Convergence in Deep Networks via the Second Law of Thermodynamics (NeurIPS 2026); a linked manuscript was unavailable when this list was checked, so reading and claims remain TBD.

Presentation focus: Derive loss decay from a PL inequality, then explain how the main paper maintains a trajectory-wise condition. Inspect its data, initialization, width and gradient-flow assumptions. Separate landscape geometry, reaching a minimum, selecting a solution and generalization. These results make concrete progress; they do not supply a general guarantee for finite deep networks trained with SGD or Adam.

Alternative focus: edge of stability and Central Flows

Background: Quadratic losses, Hessian eigenvalues and the gradient-descent stability threshold $2 / \text{learning rate}$. Cohen et al., [Gradient Descent on Neural Networks Typically Occurs at the Edge of Stability](#) (ICLR 2021), short-term loss oscillations and longer-term progress.

Main paper: Cohen, Damian, Talwalkar, Kolter & Lee, [Understanding Optimization in Deep Learning with Central Flows](#) (ICLR 2025). Start with the gradient-descent case and the authors' [illustrated explanation](#).

Recent development & discussion: Berthier, [The Multiple Timescales of Gradient Descent on the Edge of Stability: A Perturbative Derivation of the Central Flow](#) (2026), selected introduction and toy example. This formal asymptotic derivation does not provide remainder bounds or a convergence proof.

Presentation focus: Derive the quadratic stability threshold, examine self-stabilization, and show how averaging oscillations produces a smoother trajectory description. Use one example to compare the discrete iterates and central flow. Explain what this reveals about optimization dynamics without treating it as a general global-convergence theorem.

Sequence models & reasoning

State-space models and Mamba

How do recurrence, selectivity and structured matrices produce an efficient sequence model?

Background: Gu, Goel & Ré, [Efficiently Modeling Long Sequences with Structured State Spaces](#) (S4, 2021/22), selected sections on discretization and recurrence. Gu & Dao, [Mamba: Linear-Time Sequence Modeling with Selective State Spaces](#) (2023), selected sections on input-dependent dynamics.

Main paper: Dao & Gu, [Transformers Are SSMS: Generalized Models and Efficient Algorithms Through Structured State Space Duality](#) (Mamba-2, 2024).

Recent development & discussion: Lahoti et al., [Mamba-3: Improved Sequence Modeling Using State Space Principles](#) (2026).

Presentation focus: Unroll an SSM into a structured matrix and explain the restricted attention-SSM duality used by Mamba-2. Examine a state-tracking experiment from Mamba-3 and distinguish algorithmic complexity from measured hardware efficiency.

In-context learning as an algorithm

How can a forward pass perform learning without changing the network's parameters?

Background: von Oswald et al., [Transformers Learn In-Context by Gradient Descent](#) (2023), selected sections on the linear-attention construction. Review least squares, ridge regression and Bayesian prediction for linear models.

Main paper: Akyürek et al., [What Learning Algorithm Is In-Context Learning? Investigations with Linear Models](#) (2023).

Recent development & discussion: Wurgaft et al., [In-Context Learning Strategies Emerge Rationally](#) (2025).

Presentation focus: Work through one regression-learning construction and compare gradient-descent, ridge and Bayesian predictors on a controlled task distribution. Separate existence constructions and observed behavior in trained synthetic-task models from claims about general language models.

Test-time training (TTT)

What should a model learn at inference, and which objective makes those updates useful?

Scope for one session: One seminar of two academic hours. Choose learning as neural memory (shown below) or distribution-shift adaptation (alternate focus). IT3 is supplementary reading. Neither focus requires a second week.

Background: Sun, Wang, Liu, Miller, Efros & Hardt, [Test-Time Training with Self-Supervision for Generalization under Distribution Shifts](#) (ICML 2020), selected introduction and auxiliary-task formulation. Ba et al., [Using Fast Weights to Attend to the Recent Past](#) (2016), fast versus slow weights and associative memory.

Main paper: Sun et al., [Learning to \(Learn at Test Time\): RNNs with Expressive Hidden States](#) (2024). A model is the recurrent hidden state, and self-supervised gradient updates are the state transition.

Recent development & discussion: Tandon et al., [End-to-End Test-Time Training for Long Context](#) (2025). Next-token updates compress context into weights, with a meta-learned initialization. Optional adaptation comparison: Durasov, Shocher, Oner, Chechik, Efros & Fua, [IT3: Idempotent Test-Time Training](#) (ICML 2025); [implementation](#).

Presentation focus: Work through a TTT-Linear update and its recurrent computation. Compare layer-local reconstruction with end-to-end next-token adaptation, and distinguish sequence memory from adapting to a shifted test distribution. IT3 is an optional consistency-based adaptation comparison; it requires its own training setup.

Alternative focus: distribution-shift adaptation

Background: Shared encoders, auxiliary self-supervision and gradient alignment. Read selected setup and theory sections of the original Sun et al. TTT paper; the stated gradient-alignment result uses smoothness and convexity assumptions.

Main paper: Sun et al., [Test-Time Training with Self-Supervision for Generalization under Distribution Shifts](#) (ICML 2020).

Recent development & discussion: Comparison: Wang et al., [Tent: Fully Test-time Adaptation by Entropy Minimization](#) (ICLR 2021). Recent optional extension: Durasov et al., [IT3: Idempotent Test-Time Training](#) (ICML 2025), prediction feedback and consistency between successive predictions.

Presentation focus: Derive the original TTT gradient-alignment intuition and examine one adaptation experiment or failure case. Compare auxiliary self-supervision with entropy minimization; use IT3 as a short additional example if time permits. Discuss uninformative consistency, model collapse and stopping adaptation.

RL for reasoning with verifiable rewards

Does reward-based training discover successful reasoning behavior or change which existing behavior is sampled?

Background: [Spinning Up: Part 3-Introduction to Policy Optimization](#), selected sections on the log-derivative policy gradient and baselines. Reserve 25-30 minutes to introduce policies, sampled completions, rewards and group-relative advantages; prior RL coursework is not assumed.

Main paper: DeepSeek-AI et al., [DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning](#) (2025), focusing on R1-Zero and the learning objective.

Recent development & discussion: Yue et al., [Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model?](#) (2025). Counterpoint: Liu et al., [ProRL: Prolonged Reinforcement Learning Expands Reasoning Boundaries in Large Language Models](#) (2025).

Presentation focus: Derive the policy-gradient identity before introducing group-relative advantages, clipping and KL regularization. Compare pass@k evidence under the papers' different training and evaluation settings; finite sampling cannot establish that a base model assigns zero probability to a solution.

Recurrent depth and latent reasoning

Can a model gain reasoning ability by repeatedly applying the same layers?

Background: Dehghani et al., [Universal Transformers](#) (ICLR 2019; 2018 preprint). Shared layers, recurrence over depth and adaptive computation.

Main paper: Geiping et al., [Scaling up Test-Time Compute with Latent Reasoning: A Recurrent Depth Approach](#) (NeurIPS 2025).

Recent development & discussion: Jolicoeur-Martineau, [Less is More: Recursive Reasoning with Tiny Networks](#) (2025). A short comparison with recursive refinement on puzzles.

Presentation focus: Trace one recurrent block, explain how it is trained across iteration counts, and assess performance against inference compute. Use tiny recursive models as a comparison. Distinguish iteration count, parameter count and demonstrated reasoning performance.

Adaptation, representations & world models

Hypernetworks and generated adaptation

Can a network generate the parameters another network needs for a new task or context?

Background: Ha, Dai & Le, [HyperNetworks](#) (2016/17). Hu et al., [LoRA: Low-Rank Adaptation of Large Language Models](#) (2021/22), selected sections on low-rank parameter updates.

Main paper: Charakorn et al., [Text-to-LoRA: Instant Transformer Adaption](#) (2025).

Recent development & discussion: Charakorn et al., [Doc-to-LoRA: Learning to Instantly Internalize Contexts](#) (2026).

Presentation focus: Explain the conditioning signal, generated low-rank update and training objective, then examine a held-out-task or document experiment. Compare the cost and information limits of generated adaptation with fine-tuning and keeping the original context available.

JEPA and predictive world models

When does prediction in representation space support understanding, dynamics and planning?

Background: Assran et al., [Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture \(I-JEPA\)](#) (2023), selected sections on masking, target representations and collapse prevention.

Main paper: Assran et al., [V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning](#) (2025).

Recent development & discussion: Maes et al., [LeWorldModel: Stable End-to-End Joint-Embedding Predictive Architecture from Pixels](#) (2026).

Presentation focus: Trace the representation-prediction objective and action-conditioned planning loop, then analyze a planning result and its data requirements. Contrast target-network stabilization with regularizing the latent distribution; representation quality and accurate action-conditioned dynamics are separate questions.

Representation alignment and the Platonic hypothesis

Do different models learn the same representations, and how would we know?

Background: Kornblith et al., [Similarity of Neural Network Representations Revisited](#) (ICML 2019). Centered kernel alignment and the invariances of similarity measures.

Main paper: Huh et al., [Position: The Platonic Representation Hypothesis](#) (ICML 2024).

Recent development & discussion: Gröger, Wen & Brbic, [Revisiting the Platonic Representation Hypothesis: An Aristotelian View](#) (ICML 2026). Calibration for similarity scores and the distinction between global similarity and local neighborhood agreement.

Presentation focus: Compute one similarity measure on a small example, examine a cross-modal alignment result, and test its interpretation against the recent calibration critique. Treat convergence to a shared reality as a hypothesis whose evidence depends on the measurement.

Editing & interpreting networks

Model merging and task arithmetic

Why can adding parameter updates combine capabilities, and when does task interference prevent it?

Scope for one session: Choose task arithmetic or Neural Thickets for one seminar of two academic hours. Both examine capabilities near pretrained weights; averaging weights and ensembling predictions have different mechanisms and inference costs.

Background: Ilharco et al., [Editing Models with Task Arithmetic](#) (2023). Review parameter differences relative to a shared pretrained checkpoint and function-space linearization from NTK background.

Main paper: Ortiz-Jimenez, Favero & Frossard, [Task Arithmetic in the Tangent Space: Improved Editing of Pre-Trained Models](#) (2023).

Recent development & discussion: Hitit, Girrbach & Akata, [A Systematic Study of In-the-Wild Model Merging for Large Language Models](#) (2025 preprint; TMLR 2026). Alternative direction: Gan & Isola, [Neural Thickets: Diverse Task Experts Are Dense Around Pretrained Weights](#) (ICML 2026), sampling, selecting and ensembling nearby task experts.

Presentation focus: Derive how adding task vectors affects a linearized model and examine the proposed weight-disentanglement mechanism. Compare controlled task combinations with heterogeneous checkpoints, including conflicting objectives and gains relative to the best individual model.

Alternative focus: Neural Thickets and gradient-free adaptation

Background: Wortsman et al., [Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time](#) (ICML 2022), shared pretrained initialization, weight averaging and prediction ensembles. Recap task vectors from Ilharco et al., [Editing Models with Task Arithmetic](#) (2023).

Main paper: Gan & Isola, [Neural Thickets: Diverse Task Experts Are Dense Around Pretrained Weights](#) (ICML 2026). RandOpt samples Gaussian weight perturbations, evaluates candidates on task data, selects experts and ensembles their predictions. See the authors' [illustrated project](#).

Recent development & discussion: Cendra et al., [Can Bayesian Optimization Efficiently Find a Strong Single Expert in Neural Thickets?](#) (2026). A follow-up on finding one expert with fewer evaluations in the tested low-dimensional search settings.

Presentation focus: Explain expert density and diversity around a pretrained checkpoint. Work through the probability of finding a useful perturbation after repeated sampling, then compare weight averaging, expert selection and prediction ensembling. Account for task-data evaluations and ensemble inference cost. This is evidence about pretrained neighborhoods, not a proof that random initialization or unrestricted gradient descent succeeds.

Mechanistic interpretability: features and superposition

What is a feature, and can sparse autoencoders recover the units a network computes with?

Background: Elhage et al., [Toy Models of Superposition](#) (2022), selected sections on sparse features, interference and the two-feature toy model.

Main paper: Templeton et al., [Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet](#) (2024).

Recent development & discussion: Leask et al., [Sparse Autoencoders Do Not Find Canonical Units of Analysis](#) (2025).

Presentation focus: Explain the sparse-autoencoder objective and reproduce or analyze a small superposition experiment. Examine feature interventions alongside the stitching and meta-SAE evidence, asking whether interpretability, completeness and atomicity are being measured separately.

Mechanistic interpretability: circuits and causal tests

How can we trace a model's computation and test whether the proposed explanation is causal?

Background: Elhage et al., [A Mathematical Framework for Transformer Circuits](#) (2021), selected sections on the residual stream and attention's QK/OV decomposition. Review sparse representations from the features background; a separate features seminar is optional.

Main paper: Ameisen et al., [Circuit Tracing: Revealing Computational Graphs in Language Models](#) (2025).

Recent development & discussion: Kamath et al., [Tracing Attention Computation Through Feature Interactions](#) (2025).

Presentation focus: Explain how a replacement model produces an attribution graph and work through one circuit case study with an intervention. Identify approximation errors and omitted computations, then examine what the attention extension adds to the explanation.

Generative models

One-step generative models: Drifting and IMLE

How can matching real and generated samples train a generator that produces a sample in one pass?

Scope for one session: One comparative seminar of two academic hours. Use Drifting for the central derivation and IMLE / ROMS-IMLE for comparison, or choose the IMLE-first route below. Separate deeper seminars are available if both focuses are selected.

Background: Yujia Li, Swersky & Zemel, [Generative Moment Matching Networks](#) (ICML 2015), selected kernel/MMD material. Ke Li & Jitendra Malik, [Implicit Maximum Likelihood Estimation](#) (2018), the data-to-generated nearest-neighbor assignment and regression algorithm.

Main paper: Deng, He Li, Tianhong Li, Du & He, [Generative Modeling via Drifting](#) (2026). Estimate an attraction-repulsion field from real and generated samples and fit frozen drifted targets.

Recent development & discussion: Chirag Vashist & Ke Li, [ROMS-IMLE: A Minimalist Approach to Competitive Single-Step Generative Modelling](#) (2026). Compare nearest-neighbor assignment, robust losses and intermediate-resolution supervision with the drifting update. Optional theory: Esteban-Casadevall et al., [Kernel-Gradient Drifting Models](#) (2026), including Max Welling.

Presentation focus: Introduce sample-based distribution matching, derive the drifting field and frozen-target regression, and study one toy experiment. Compare IMLE's data-to-generated matching and ROMS-IMLE's practical changes briefly. Choose one main mechanism to derive; keep zero-drift and maximum-likelihood claims tied to their assumptions.

Alternative focus: IMLE and mode coverage

Background: Ke Li & Jitendra Malik, [Implicit Maximum Likelihood Estimation](#) (2018). Implicit distributions, nearest-neighbor assignment, mode coverage and the likelihood connection under stated assumptions.

Main paper: Chirag Vashist & Ke Li, [ROMS-IMLE: A Minimalist Approach to Competitive Single-Step Generative Modelling](#) (2026). Revisit IMLE with intermediate-resolution supervision and robust losses.

Recent development & discussion: ROMS-IMLE is the recent anchor. The authors' [method page](#) provides additional explanation and experiments. Comparison: Deng et al., [Generative Modeling via Drifting](#) (2026), selected algorithm and toy example.

Presentation focus: Explain the direction of nearest-neighbor matching, latent assignment and alternating regression, then examine sampling cost and the ROMS-IMLE ablations. Keep the maximum-likelihood connection tied to its assumptions. Use Drifting as a brief comparison within the same two-academic-hour seminar.

Diffusion language models

How can denoising generate discrete text, and what changes when generation need not proceed left to right?

Background: Austin et al., [Structured Denoising Diffusion Models in Discrete State-Spaces](#) (2021), selected sections on transition matrices and absorbing-mask corruption.

Main paper: Sahoo et al., [Simple and Effective Masked Diffusion Language Models](#) (2024).

Recent development & discussion: Nie et al., [Large Language Diffusion Models](#) (LLaDA, 2025).

Presentation focus: Derive the masked-denoising objective and trace one sampling schedule, then compare refinement, parallelism and likelihood evaluation with autoregressive generation. The topic requires discrete corruption and language-specific evaluation beyond a continuous diffusion refresher.

Graphs, symmetry & neural architectures

Symmetry, equivariance and canonicalization

Can we enforce symmetry by putting each input into a canonical orientation?

Scope for one session: Advanced mathematical candidate. A concrete rotation or permutation example is enough to motivate the main result within two academic hours.

Background: Cohen & Welling, [Group Equivariant Convolutional Networks](#) (ICML 2016). Group actions, equivariance and weight sharing.

Main paper: Dym, Lawrence & Siegel, [Equivariant Frames and the Impossibility of Continuous Canonicalization](#) (ICML 2024).

Recent development & discussion: Shumaylov et al., [Lie Algebra Canonicalization: Equivariant Neural Operators under arbitrary Lie Groups](#) (ICLR 2025; 2024 preprint). Selected introduction and one application.

Presentation focus: Start with a group-action example, show why a canonical choice can become discontinuous under the stated assumptions, and explain how weighted frames address continuity. Leave general topology proofs and non-compact Lie-group machinery for optional reading.

Oscillatory neurons and synchronization

Can neural representations bind information through synchronization?

Scope for one session: Exploratory candidate connecting neural computation with dynamical systems, including work by Max Welling.

Background: Strogatz, [From Kuramoto to Crawford: exploring the onset of synchronization in populations of coupled oscillators](#) (2000). Sections 2 and 3.1-3.3 on coupled phases and synchronization (author-hosted PDF).

Main paper: Miyato, Löwe, Geiger & Welling, [Artificial Kuramoto Oscillatory Neurons](#) (ICLR 2025; 2024 preprint).

Recent development & discussion: Iqbal et al., [Spontaneous symmetry breaking and Goldstone modes for deep information propagation](#) (2026 preprint). A related perspective on symmetry and persistent information propagation, rather than a direct analysis of AKOrN.

Presentation focus: Derive the two-oscillator intuition, trace the generalized neural update, and study one object-discovery or reasoning experiment. Use the Goldstone paper for a brief connection to depth and memory; its physics derivations are optional.

KANs: learnable functions and scientific discovery

When does learning the functions on network edges help more than learning ordinary weights?

Scope for one session: Two academic hours: the original architecture, one scientific example and one controlled comparison. Approximation-rate proofs are optional.

Background: Kolmogorov, [On the Representation of Continuous Functions of Several Variables as Superpositions of Continuous Functions of One Variable and Addition](#) (1957), theorem statement only. Review one-dimensional spline bases using the main paper's architecture section. Separate representation existence from efficient approximation and learnability.

Main paper: Liu et al., [KAN: Kolmogorov-Arnold Networks](#) (ICLR 2025; 2024 preprint), selected sections on learned edge functions, grid refinement and one function-fitting experiment.

Recent development & discussion: Liu et al., [KAN 2.0: Kolmogorov-Arnold Networks Meet Science](#) (2024), one scientific example. Essential comparison: Yu, Yu & Wang, [KAN or MLP: A Fairer Comparison](#) (2024). Recent optional theory: Kratsios, Kim & Furuya, [Approximation Rates in Besov Norms and Sample-Complexity of Kolmogorov-Arnold Networks with Residual Connections](#) (2025), assumptions and result statements.

Presentation focus: Build a small spline-based KAN and explain its compositional bias and possible formula recovery. Compare against an MLP with matched parameters or FLOPs and a spline-equipped MLP. Ask which benefit comes from the function basis and which comes from the architecture; the representation theorem alone does not establish a practical advantage.

GNNs: message passing and graph transformers

What graph structure can a neural network distinguish, and what does global attention add?

Scope for one session: Two academic hours on message-passing expressivity and structural encodings in graph transformers.

Over-squashing is a separate alternative focus, rather than another required component.

Background: Sanchez-Lengeling et al., [A Gentle Introduction to Graph Neural Networks](#) (Distill 2021), a visual primer. Xu, Hu, Leskovec & Jegelka, [How Powerful are Graph Neural Networks?](#) (ICLR 2019), multiset aggregation, GIN and the 1-Weisfeiler-Leman expressivity ceiling.

Main paper: Stoll, Müller & Morris, [Generalizable Insights for Graph Transformers in Theory and Practice](#) (NeurIPS 2025), selected theory sections and one structural-encoding ablation.

Recent development & discussion: Ma et al., [Plain Transformers Can be Powerful Graph Learners](#) (TMLR 2026; 2025 preprint). Examine the graph-specific adaptations behind the title. Optional benchmark discussion: Bechler-Speicher et al., [Position: Graph Learning Will Lose Relevance Due To Poor Benchmarks](#) (ICML 2025).

Presentation focus: Work through a pair of graphs that 1-WL-style message passing cannot distinguish. Explain how structural or positional information changes the comparison with a graph transformer. Distinguish theoretical expressivity, measured accuracy and benchmark quality; leave the full higher-order hierarchy for optional reading.

Alternative focus: over-squashing and graph geometry

Background: Kipf & Welling, [Semi-Supervised Classification with Graph Convolutional Networks](#) (ICLR 2017), neighborhood propagation. Alon & Yahav, [On the Bottleneck of Graph Neural Networks and its Practical Implications](#) (ICLR 2021), information bottlenecks and over-squashing.

Main paper: Wilson, Bechler-Speicher & Velickovic, [Cayley Graph Propagation](#) (2025), an expander-based propagation template.

Recent development & discussion: Liang et al., [Mitigating Over-Squashing in Graph Neural Networks by Spectrum-Preserving Sparsification](#) (ICML 2025), a comparison of graph modification and propagation costs.

Presentation focus: Use a bottleneck graph to distinguish over-squashing from over-smoothing. Explain expander connectivity intuitively, then inspect one propagation experiment and the cost of modifying graph connectivity. A complete group-theory or spectral proof is optional.

Course website: [Deep Deep Learning](#)